WORKED SOLUTION

STEP II 2021, Question 7

Pure • Rotation Matrices

STEP II 2021, Question 7

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

A 2×2 rotation matrix has a very rigid shape: it is $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, so its columns are unit vectors. Asking that $\mathbf{R} + \mathbf{I}$ keeps this shape pins down ϕ completely. For a general matrix with $\mathbf{S}^3 = \mathbf{I}$, the useful tools are the determinant (which is multiplicative) and the relation $\mathbf{S}^2 = (a+d)\mathbf{S} - \mathbf{I}$, which lets us write any power of $\mathbf{S}$ as a combination of $\mathbf{S}$ and $\mathbf{I}$.

Solution

Part (i)

Write $c = \cos\phi$ and $s = \sin\phi$, so

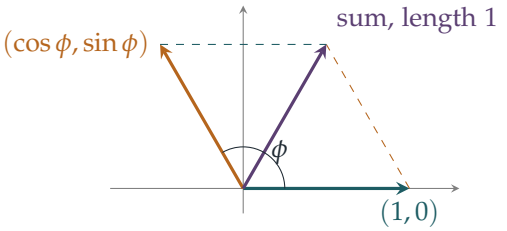
$$\mathbf{R} = \begin{pmatrix} c & -s \\ s & c \end{pmatrix}, \qquad \mathbf{R} + \mathbf{I} = \begin{pmatrix} 1+c & -s \\ s & 1+c \end{pmatrix}.$$

This already has the pattern $\begin{pmatrix} p & -q \\ q & p \end{pmatrix}$, and such a matrix is a rotation exactly when $p^2 + q^2 = 1$ (then $(p, q) = (\cos\theta, \sin\theta)$ for some θ). So we need

$$(1+c)^2 + s^2 = 1 \iff 2 + 2c = 1 \iff \cos\phi = -\tfrac{1}{2}.$$

With $0^\circ \leq \phi < 360^\circ$ this gives $\phi = 120^\circ$ or $\phi = 240^\circ$.

Geometrically, the first column of $\mathbf{R} + \mathbf{I}$ is the sum of the unit vectors $(1, 0)$ and $(\cos\phi, \sin\phi)$, and we need this sum to be a unit vector too: the three vectors form an equilateral triangle, as in the picture for $\phi = 120^\circ$.



In either case $\mathbf{R}^3$ is a rotation through 3ϕ, which is $360°$ or $720°$, a whole number of turns. Hence $\mathbf{R}^3 = \mathbf{I}$.

Part (ii)

The determinant. Since det is multiplicative, $(\det \mathbf{S})^3 = \det(\mathbf{S}^3) = \det \mathbf{I} = 1$. The matrix is real, so $\det \mathbf{S}$ is a real number whose cube is 1, and the only such number is 1. So $\det \mathbf{S} = ad - bc = 1$.

The quadratic relation. Multiplying out,

$$\mathbf{S}^2 = \begin{pmatrix} a^2 + bc & ab + bd \\ ca + dc & bc + d^2 \end{pmatrix}, \qquad (a+d)\mathbf{S} - \mathbf{I} = \begin{pmatrix} a^2 + ad - 1 & ab + bd \\ ac + dc & ad + d^2 - 1 \end{pmatrix}.$$

The off-diagonal entries agree, and the diagonal entries agree because $bc = ad - 1$ (that is, $\det \mathbf{S} = 1$). Hence $\mathbf{S}^2 = (a+d)\mathbf{S} - \mathbf{I}$, as required.

The trace. Let $t = a + d$. Multiply the relation by $\mathbf{S}$ and use it again:

$$\mathbf{S}^3 = t\mathbf{S}^2 - \mathbf{S} = t(t\mathbf{S} - \mathbf{I}) - \mathbf{S} = (t^2 - 1)\mathbf{S} - t\mathbf{I}.$$

Setting $\mathbf{S}^3 = \mathbf{I}$ gives

$$(t^2 - 1)\mathbf{S} = (t+1)\mathbf{I}.$$

? **Ask yourself** Can we just divide by $t^2 - 1$ and be done?

Not without care: we must look at the case $t^2 - 1 = 0$ separately, and in fact it is the case that happens.

- If $t^2 \neq 1$, then $\mathbf{S} = \dfrac{t+1}{t^2-1}\mathbf{I} = k\mathbf{I}$ with k real. Then $\mathbf{S}^3 = k^3\mathbf{I} = \mathbf{I}$ forces $k^3 = 1$, so $k = 1$ and $\mathbf{S} = \mathbf{I}$, which is excluded.
- So $t^2 = 1$. The equation becomes $\mathbf{0} = (t+1)\mathbf{I}$, so $t = -1$ (the value $t = 1$ would give $2\mathbf{I} = \mathbf{0}$, which is false).

Therefore $a + d = -1$.

Part (iii)

First, $\mathbf{S} \neq \mathbf{I}$: otherwise $\mathbf{S} + \mathbf{I} = 2\mathbf{I}$, which has determinant 4 and doubles lengths, so is not a rotation. Part (ii) therefore applies, and $a + d = -1$.

Since $\mathbf{S}+\mathbf{I}$ is a rotation, say through θ,

$$\mathbf{S}=\begin{pmatrix}\cos\theta-1 & -\sin\theta\\ \sin\theta & \cos\theta-1\end{pmatrix}.$$

The trace condition gives $2\cos\theta-2=-1$, so $\cos\theta=\frac{1}{2}$ and $\sin\theta=\pm\frac{\sqrt{3}}{2}$. Then

$$\mathbf{S}=\begin{pmatrix}-\frac{1}{2} & -\sin\theta\\ \sin\theta & -\frac{1}{2}\end{pmatrix},$$

which has the rotation pattern with $\left(-\frac{1}{2}\right)^2+\sin^2\theta=\frac{1}{4}+\frac{3}{4}=1$. So $\mathbf{S}$ is a rotation, with cosine $-\frac{1}{2}$ and sine $\pm\frac{\sqrt{3}}{2}$.

The angle is 120° (when $\theta=60^\circ$) or 240° (when $\theta=300^\circ$). Both really occur: by part (i), each of these rotations has $\mathbf{S}+\mathbf{I}$ a rotation and $\mathbf{S}^3=\mathbf{I}$. (Equivalently, once we know $\mathbf{S}$ is a rotation, part (i) itself lists the possible angles.)

Checks

- With $\phi=120^\circ$, $\mathbf{R}+\mathbf{I}=\begin{pmatrix}\frac{1}{2} & -\frac{\sqrt{3}}{2}\\ \frac{\sqrt{3}}{2} & \frac{1}{2}\end{pmatrix}$, which is the rotation through 60°; with $\phi=240^\circ$ it is the rotation through 300°.
- The matrix $\mathbf{S}=\begin{pmatrix}0 & -1\\ 1 & -1\end{pmatrix}$ has $\mathbf{S}^3=\mathbf{I}$ (multiply it out) and is not a rotation; it still has determinant 1 and trace -1, as part (ii) predicts. Its $\mathbf{S}+\mathbf{I}=\begin{pmatrix}1 & -1\\ 1 & 0\end{pmatrix}$ is not a rotation, consistent with part (iii).
- A computer check of many random real matrices with $\mathbf{S}^3=\mathbf{I}$, $\mathbf{S}\neq\mathbf{I}$ confirmed determinant 1, trace -1 and the relation for $\mathbf{S}^2$ every time; scanning θ in part (iii) found only $\theta=60^\circ,300^\circ$.

Common slips

- Dividing by t^2-1 without checking whether it can be zero; the case $t^2=1$ is exactly the answer.
- Forgetting that $\mathbf{S}\neq\mathbf{I}$ must be established in part (iii) before part (ii) can be used.
- Thinking $\det(\mathbf{S}+\mathbf{I})=1$ is enough to make $\mathbf{S}+\mathbf{I}$ a rotation; here it holds automatically, so it gives no information.

Beyond STEP

The relation $\mathbf{S}^2=(a+d)\mathbf{S}-(\det\mathbf{S})\mathbf{I}$ is the 2×2 case of the Cayley–Hamilton theorem: every square matrix satisfies its own characteristic polynomial. Part (ii) says that a real

2×2 matrix of order 3 has eigenvalues the two non-real cube roots of unity, and such a matrix is always *similar* to a rotation through $120°$, though not necessarily equal to one. Strang's *Introduction to Linear Algebra* develops eigenvalues and similarity.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.